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Radical Addition

Regioselective Intramolecular Allene Amidation Enabled by an EDA Complex**

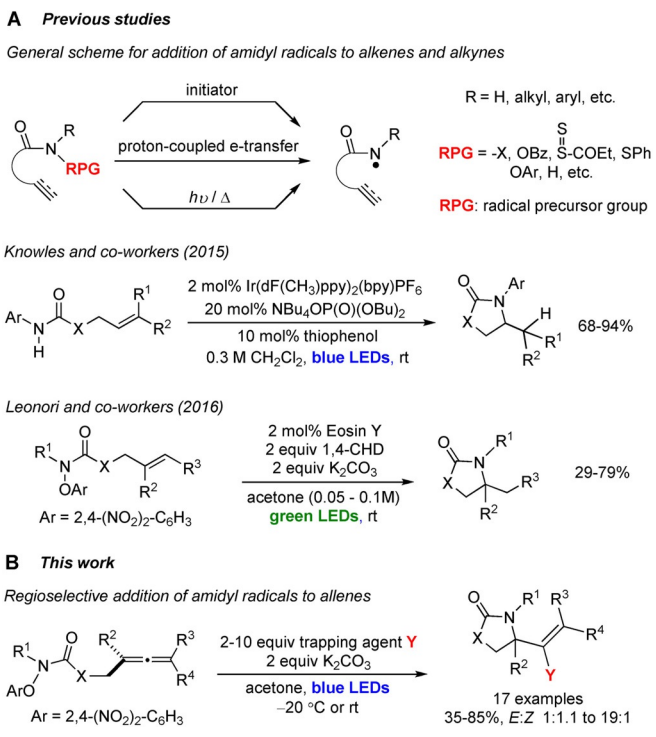
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Abstract: The addition of radicals to unsaturated precursors is a powerful tool for the synthesis of both carbo- and heterocyclic organic building blocks. The recent advent of mild ways to generate N-centered radicals has reignited interest in exploiting highly regio-, chemo-, and stereoselective transformations that employ these reactive intermediates. While the additions of aminyl, iminyl, and amidyl radicals to alkenes and alkynes have been well-studied, analogous additions to allenes are scarce. Allenes offer several attractive features, including potential for selective amidation at three distinct sites via judicious choice of precursor or radical source, the opportunity for axial-to-point chirality transfer, and productive trapping of vinyl or allyl radical intermediates to diversify functionality in the products. In this article, we report a regioselective addition of amidyl radicals to allenes to furnish an array of valuable N-heterocycle scaffolds.

Since the first report of the triphenylmethyl radical by Moses Gomberg in 1900, reactions involving radical intermediates have become important tools for the efficient functionalization of organic molecules.[1] A recent renaissance in radical chemistry has resulted in several mild strategies to generate and exploit these reactive intermediates, including N-centered radicals.[2] While the additions of N-radicals to alkenes and alkynes have been well-studied, the analogous regio- and stereo-controlled additions of these species to allenes are underdeveloped.[3] This is despite the potential for the expedient construction of N-heterocycles that appear in pharmaceuticals, agrochemicals, natural products, and materials.

Allenes contain two cumulated, orthogonal π-bonds and three distinct sites for potentially tunable addition of a N-radical species. Allenes also typically display higher reactivity than their alkene counterparts, as well as greater thermodynamic stability compared to the corresponding alkynes. Most interestingly, radical addition to a allene can occur at the proximal α carbon (Scheme 1 B) to generate a vinyl radical, at the β carbon to furnish an allyl radical, or at the other γ carbon to yield an endocyclic vinyl radical. While formation of the stabilized allyl radical might be expected to be preferred, the orthogonality of the two π-systems means that stabilization of this radical requires a 90° rotation; this additional energy requirement can often result in the favorable kinetic formation of a vinyl radical instead.[3a,4] This vinyl radical can be trapped with a variety of electrophiles to introduce additional functionality into the products or participate downstream in radical cascade reactions.

Nitrogen-centered radicals come in a variety of "flavors", with the overall reactivities broadly categorized as nucleophilic, electrophilic, or ambiphilic.[2b] Amidyl radicals are popular for their umpolung reactivity and typically display electrophilic behavior in their preferred reactions with electron-rich olefins. A variety of novel strategies to generate amidyl radicals have been developed over the last decade; the corresponding cyclization reactions with alkene or alkyne coupling partners have



Scheme 1. Previous studies and exploring the factors that influence the site-selectivity of *N*-radical additions to allenes.

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[**] *EDA complex = Electron Donor–Acceptor complex*

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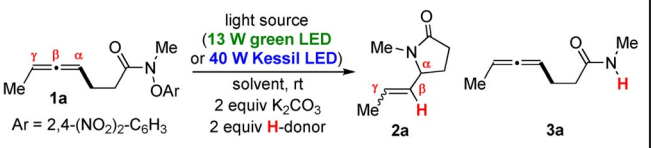
been demonstrated under photochemical,[5] electrochemical,[6] and chemical conditions[7] (Scheme 1 A). Selected examples include recent work by the groups of Knowles and Leonori, where the use of either a metal or organic photocatalyst furnishes γ-lactam analogs and other N-heterocycles.[5a,c]

Our laboratory's interest in new methods for stereocontrolled C–N bond formation, coupled with the potential of allenes as radical acceptors, prompted us to investigate whether regioselective amidyl radical addition to allenes could be achieved with concomitant modification to all three unsaturated carbons of the allene. Preliminary studies explored the influence of the precursor substitution pattern, the tether length between the nitrogen and the allene, the electronic features of the radical, and the reaction conditions on the regioselectivity of addition of the N-centered radical to the allene. Establishing these general trends will ultimately enable these reactivities to be harnessed to furnish an array of powerful scaffolds for the syntheses of valuable N-heterocycles (Scheme 1 B) that are complementary to transition metal and ionic-based protocols. Herein, we report a mild, catalyst-free addition of amidyl radicals to allenes that proceeds with high regioselectivity for addition to the α site to generate intermediate vinyl radicals that can be trapped with diverse electrophiles.

We began our investigations employing electrophilic amidyl radicals, which to the best of our knowledge have not been studied in terms of their reactivity towards allenes. These radicals are typically generated via functionalization of an amide nitrogen with a radical precursor group (RPG), followed by homolytic cleavage of the *N*-RPG bond in the presence of a radical initiator, heat, irradiation, and/or a photocatalyst (PC).[5–7] Photochemical generation of amidyl radicals is particularly attractive, as this strategy avoids the use of toxic initiators and harsh reaction conditions.

The Leonori group has popularized aryloxyamides as precursors to amidyl radicals using Eosin Y as the photocatalyst (Scheme 1 A). This strategy is attractive, as it avoids the use of rare noble metals and strong oxidants that might engage in undesired reaction with the allene moiety.[8] Inspired by this work, aryloxyamide **1a** was prepared and subjected it to visible light irradiation (Table 1). Irradiation of **1a** using a 13 W green LED in the presence of Eosin Y, K_2CO_3, and 1,4-cyclohexadiene (1,4-CHD) as the H-atom donor in acetone furnished 41% of the 5-*exo*-trig cyclization product **2a** as a mixture of stereoisomers (entry 1). To our surprise, 18% of **2a** was observed even in the absence of Eosin Y (entry 2). While the UV-vis spectrum of **1a** indicates a $\lambda_{max} < 350$ nm, the fact that irradiation with green LED and CFL (see the Supporting Information for details) furnishes **2a** prompted us to try other light sources to improve yields without the need for Eosin Y. Irradiation at 440 nm (see the Supporting Information for wavelengths screened) significantly improved the yield of **2a** to 78% (entry 4). While dilution from 0.1 M in acetone (entry 3) to 0.025 M (entry 4) did not change the yield or *E*:*Z* ratio, it did shorten the reaction time. Presumably, decrease reaction time is the result of increase transparency of the reaction mixture. DMSO and MeCN (entry 5) were also effective solvents (see the Supporting Information for a complete solvent screen). The

Table 1. Optimization of amidyl radical addition to allenes.



entry	source	solvent	H-donor	time [h]	% yield[a]	*E*:*Z*
1[b]	Green LED	0.1 M acetone	1,4-CHD	21	41	2.1:1
2	Green LED	0.1 M acetone	1,4-CHD	21	18	2.2:1
3	440 nm	0.1 M acetone	1,4-CHD	5	76	2.2:1
4	440 nm	0.025 M acetone	1,4-CHD	1.5	78	2.1:1
5	440 nm	0.025 M MeCN	1,4-CHD	3	71	2.1:1
6	440 nm	0.025 M acetone	9,10-DHA	4	71	2.1:1
7	440 nm	0.025 M acetone	Hantzsch ester	1	74	2.3:1
8	no light	0.025 M acetone	1,4-CHD	1.5	0	–
9	440 nm	0.025 M acetone	1,4-CHD[c]	24	0	–
10[d]	440 nm	0.025 M acetone[e]	1,4-CHD	24	49	2:1
11	440 nm	0.025 M acetone	1,4-CHD	16	0	–
12[f]	440 nm	0.025 M acetone	1,4-CHD	18	85	3.2:1
13[g]	440 nm	0.025 M acetone	1,4-CHD	92	83	3.6:1

[a] ^{1}H NMR yield of *E*:*Z* isomers with 1,3,5-trimethoxybenzene as internal standard. [b] 2 mol% Eosin Y. [c] no K_2CO_3. [d] [D_6]acetone solvent. [e] 3:1 D:H ratio. [f] −20 °C. [g] −40 °C.

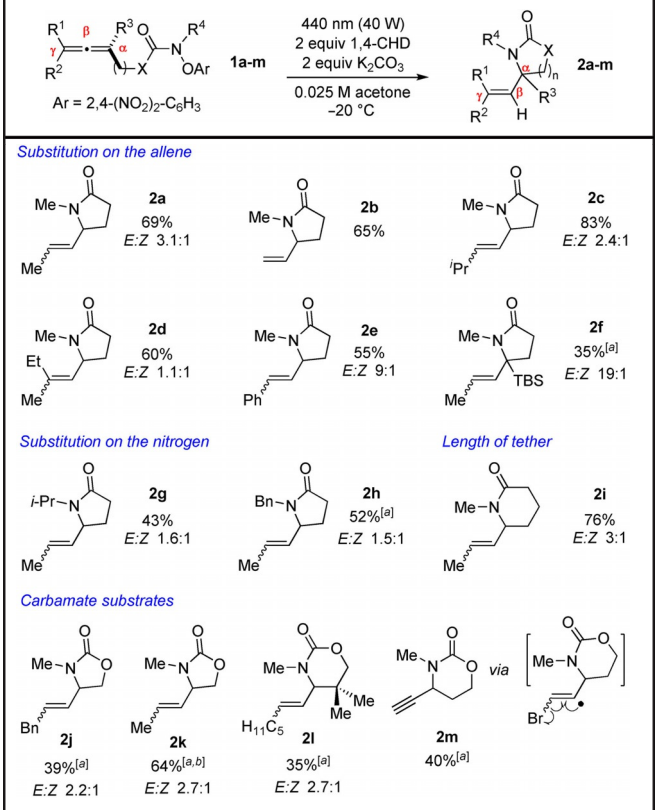
1,4-CHD could be replaced with other hydrogen atom donors, including 9,10-dihydroanthracene (9,10-DHA) and Hantzsch ester (entries 6–7). Attempts to increase the *E*:*Z* ratio by employing bulky H-donors delivered lower yields with no increase in stereoselectivity (see Supporting Information for complete list of H-donors screened).

Control experiments (entries 8–9) demonstrate cyclization does not proceed in the absence of light or K_2CO_3. Interestingly, carrying out the reaction in acetone-d_6 in the absence of 1,4-CHD (entry 10) gave 49% of **2a**, with 37% showing D-incorporation from the solvent. As expected, removing both 1,4-CHD and K_2CO_3 from the mixture (entry 11) resulted in no reaction to **2a**. Decreasing the temperature to −20 °C (entry 12) and −40 °C (entry 13) led to less competing reduction of the amidyl radical to **3a**[9] and improved both yield and *E*:*Z* ratio. Notably, no H-atom abstraction product **3a** was observed at the lower reaction temperatures.

With optimized conditions in hand, the scope of the reaction was investigated (Table 2). Several substrates with a three-carbon tether between the nitrogen and allene were investigated. In these cases, 5-*exo*-trig cyclization is expected to be favored over competing 6-*endo*-dig cyclization, due to stereoelectronic prerequisites for radical addition to a π-bond.[10] Allene **1a** delivered **2a** in 69% isolated yield, while irradiating monosubstituted allene **1b** gave **2b** in 65% yield. Allene **1c**, containing an isopropyl group at the γ-carbon, was prepared to assess whether increased $A_{1,3}$-strain in the vinyl radical intermediate would improve the *E*:*Z* ratio. Irradiation under the reaction conditions resulted in a similar *E*:*Z* ratio of 2.4:1 in **2c**; however, a 1-Me-3-*i*Pr-trisubstituted allene gave no cyclized

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Table 2. Scope of intramolecular addition of *N*-radicals to allenes.



[a] Reactions were carried out at rt. [b] ^{1}H NMR yield using 1,3,5-trimethoxybenzene as internal standard.

product (see the SI for details). When R^1 and R^2 are of similar size, as in **1d**, the *E*:*Z* ratio dropped to 1.1:1 in **2d**.

We hypothesized the *E*:*Z* ratio could be improved by installing either an aryl group at the γ site of the allene or a bulky group at the α carbon. Irradiation of allene **1e** gave 55% of **2e** with a *E*:*Z* ratio of 9:1. The increase in stereoselectivity is attributed to unfavorable electron-electron repulsion between the radical and the π-system of the aromatic ring.[11] Selectivity was improved further when a TBS group was present at the α carbon in **1f**, giving a 19:1 *E*:*Z* ratio in **2f**, albeit in 35% yield. The higher selectivity observed in **1f** is attributed to $A_{1,3}$-strain of the vinyl radical intermediate.[12]

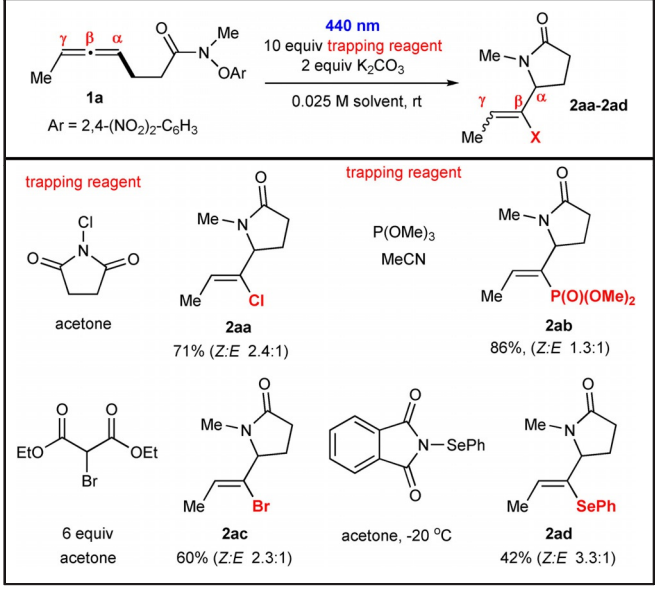
Substitution at the nitrogen was explored with substrates **1g,h** (Table 2), which have the potential of undergoing further HAT or radical cyclization. Unfortunately, no radical cascade products were evident in the reaction mixture and **2g,h** were obtained in slightly lower yields and *E*:*Z* ratios as compared to **2a**. This is consistent with previous reports that demonstrate increased bulk at the nitrogen results in more reduction product at the expense of the desired cyclization (Table 1).[13]

The influence of the tether length between the allene and the amidyl radical was explored (Table 2). When $n=0$ and $X=CH_2$ in **1**, only H-abstraction was noted, presumably due to the reversible formation of the vinyl radical intermediate (see the Supporting Information for details).[14] When $n=2$ and $X=CH_2$ in **1i**, 6-*exo*-trig cyclization produces **2i** in 76% yield with trace amounts of H-atom abstraction product. This is notable compared to previous reports of 1,5-HAT in analogous amidyl radical cyclizations with alkenes. Thus, the higher reactivity of allenes can be attributed to reduced formation of the H-atom abstraction by-product.

Carbamates were tolerated under the reactions conditions and led to 1,2- and 1,3-cyclic carbamates **2j–2l**. In agreement with previous reports, lower yields of cyclization was observed due to increased electrophilicity of the resulting amidyl radical.[5a] However, employing carbamates enabled the synthesis and cyclization of substrates with substitution patterns that were difficult to access with traditional allene chemistry, such as **1m**. Cyclization of **1m** furnished **2m** in a 40% yield; the alkyne likely forms from radical elimination of bromine (**1m′**).

The intermediate vinyl radical could be trapped with other electrophiles in addition to hydrogen atom donors. Employing an excess of the radical trapping agents *N*-chlorosuccinimide (NCS), trimethoxyphospine, diethyl bromomalonate, and *N*-(phenylseleno)phthalimide furnished the functionalized alkenes **2aa–2ad** in good yields (Table 3). Notably, the use of NCS resulted in a 71% yield of vinyl chloride **2aa**, with no **2a** detected in the crude ^{1}H NMR. Bromomalonate was superior to *N*-bromosuccimide as a Br-donor. While the equivalents of the trapping agent could be reduced to 3 equiv without a significant decrease in the yield of **2ac**, a greater amount of **2a** was observed at lower loadings. Cyclization of **1a** at −20 °C gave **2ad** in 69% yield and 3.3:1 *E*:*Z* ratio. Unfortunately, other intermolecular trapping agents, including Michael acceptors, O_2 and various heteroatom donors resulted only in **2a** or decomposition. Mechanistic investigations were undertaken to understand several unique features of this chemistry, including the reasons why no photocatalyst is required, the effect of different salts, and whether the reaction proceeds through: 1) Direct cleavage of the N–O bond, 2) an exciplex complex involving the allene, or 3) an electron donor–acceptor (EDA) complex.

Table 3. Intermolecular trapping of the vinyl radical intermediates.



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While a photocatalyst is not required, we were curious as to why K_2CO_3 is key to successful reaction. Initial insights into the role of this additive were carried out by varying both the anion and cation of the salt (selected results in Table 4). KCl (entry 1) gave lower reactivity and yield, while $KHCO_3$ (entry 2) and other potassium salts (see Supporting Information for details) performed similarly to K_2CO_3, albeit with prolonged reaction times. The lack of a clear trend between the salt's basicity and yield and/or reaction time argues against it serving solely as a Brønsted base. Cs (entry 5) and Li (entry 6) carbonates gave similar results to K_2CO_3, although Na_2CO_3 gave a lower yield (entry 7). The higher solubility of Cs_2CO_3 and tetrabutylammonium acetate (entries 5, 8) reduced the reaction time compared to K_2CO_3. Finally, decreasing the K_2CO_3 to 0.2 equivalents (entry 9) still gave 83% yield of **2a** after 5.6 h. Taken together, these results cannot rule out the role of the salt as a simple electrolyte, where irradiation at 440 nm directly excited **1a** to promote either direct homolysis of the N–O bond or a SET event that initiates formation of the amidyl radical.

The intermediacy of radical species was supported by adding TEMPO to the reaction to give the trapped product **2a-T**, as detected by both mass spectrometry and ^{1}H NMR analysis of the crude reaction (Scheme 2). The fact that 14% of **1a** remains after 1.5 h of irradiation argues against direct breaking of the weak N–O bond, as complete cleavage would be expected in the presence of TEMPO under the reaction conditions.

Another possibility involves formation of an exciplex complex between the allene and the RPG, as similar mechanisms have been proposed in other radical additions to allenes.[15] To rule out the likehood of an allene exciplex complex, the amidyl radical precusor **6** was employed (Scheme 3). Under the reaction conditions, the H-atom abstraction product **7** was observed, providing evidence against an exciplex complex involving the allene as necessary for the N–O bond cleavage.

Finally, UV-vis experiments were carried out to assess the potential for formation of an electron donor-acceptor (EDA) complex. A recent report by Pericàs described a EDA complex between K_2CO_3 and an electron-poor aryl group, based on the changes in absorption of different components.[8e] In our system, a slight difference in the UV/Vis spectra of **1a** in acetone and **1a**/K_2CO_3 in acetone (red and black traces, respectively, Figure 1) was observed. In contrast, when K_2SO_4 was used as the donor, no change in absorbance between **1a** in acetone and **1a**/K_2SO_4 in acetone was noted (see the SI for more details). Indeed, employing K_2SO_4 as the donor instead of K_2CO_3 under optimized conditions (Table 4, entry 3) gave no cyclized product **2a**.

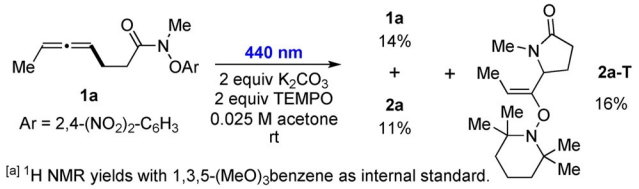
A mechanism was proposed based on our preliminary findings (Scheme 4). An EDA complex forms from **1a** and the donor K_2CO_3 (D). Irradiation of **A** results in a SET event to yield **B**, which fragments to furnish **C** and **D**. Three possible fates of **C/D** include: (1) Direct reduction of **D** to **3a**; (2) 5-*exo*-trig cyclization to give *E*- and *Z*-isomers of intermediate **E**, or (3) dimerization of the amidyl radical (not detected). Trapping of intermediates **E-(*Z*)** and **E-(*E*)** furnishes **2a**. A catalytic amount of K_2CO_3 can be utilized, as it is proposed to be turned over by a SET event with the radical from H-atom abstraction from 1,4-CHD or with an aryloxy fragment. Finally, a lights on/off experiment indicated negligible conversion when the light was off and no isomerization of **2a-(*E*/*Z*)** once formed (see the Supporting Information for details).

Table 4. Investigation into the effect of the salt in the reaction.



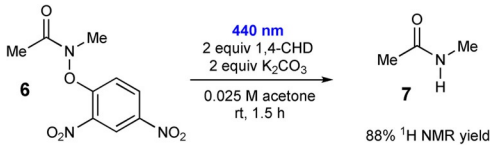
entry	salt	time [h]	**2a** (%)	*E*:*Z*	**1a** (%)
1	KCl	25	62	2.1:1	16
2	$KHCO_3$	9	76	2.2:1	0
3	K_2SO_4	24	0	–	64
4	KO^{t}Bu	1.2	23	2.2:1	0
5	Cs_2CO_3	50 min	72	2.3:1	0
6	Li_2CO_3	72	60	2.1:1	10
7	Na_2CO_3	5	32	2.3:1	9
8	TBAOAc	2	65	2.2:1	0
9[b]	K_2CO_3	5.6	83	2.3:1	0

[a] ^{1}H NMR yield using 1,3,5-trimethoxybenzene as the internal standard. [b] 0.2 equivalent of K_2CO_3 was used.



[a] ^{1}H NMR yields with 1,3,5-(MeO)$_3$benzene as internal standard.

Scheme 2. Employing TEMPO has a radical trapping agent.



Scheme 3. Cleavage of the N–O bond in the absence of an allene.

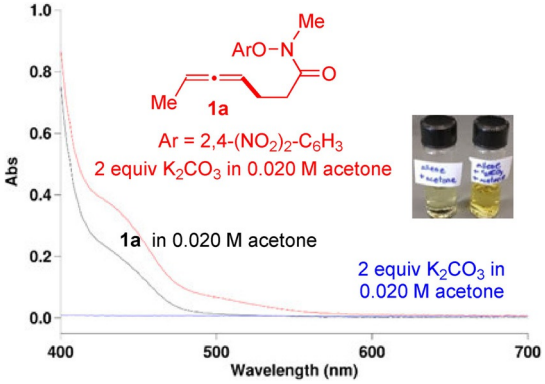
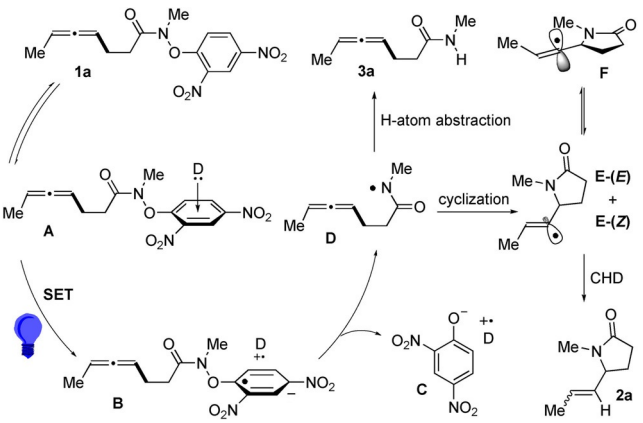


Figure 1. UV-vis plots of K_2CO_3 (blue), **1a** (black) and **1a**/K_2CO_3 (red) in acetone.

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Scheme 4. Mechanistic proposal.

In conclusion, we have developed a direct and regioselective addition of amidyl radicals to allenes to furnish highly modifiable γ-lactams that can serve as scaffolds for drugs and natural products. Various substitution patterns are well-tolerated and diverse electrophiles can be used to trap the intermediate vinyl radicals. Ongoing investigations are targeted towards methods that deliver products where all three allene carbons are functionalized, as well as experimental and computational studies of other radical additions to allenes to achieve complexity-forming cascade reactions.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: amidyl radicals · EDA complexes · lactams · radical additions to allenes · vinyl radicals

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